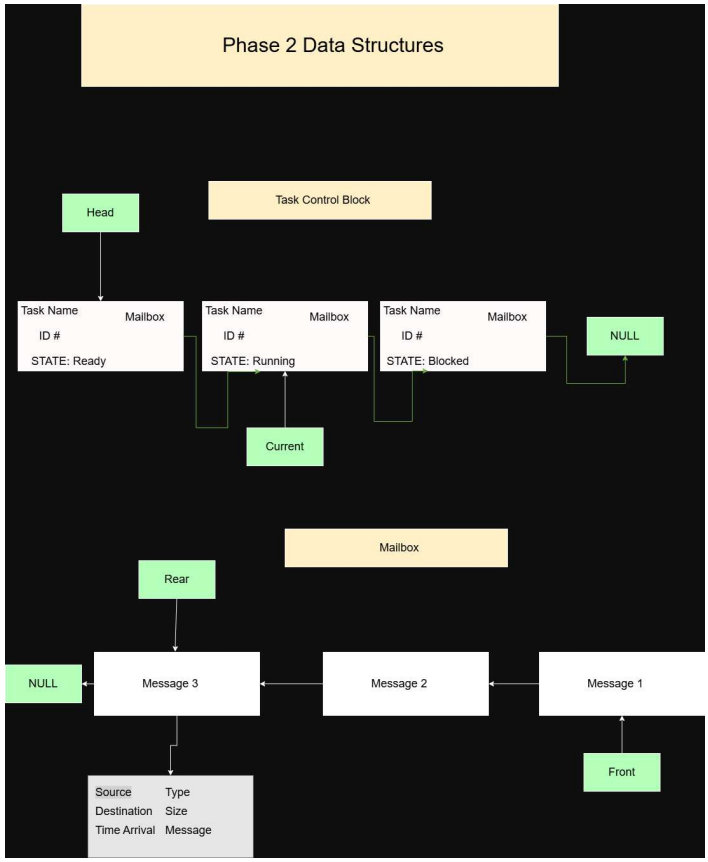
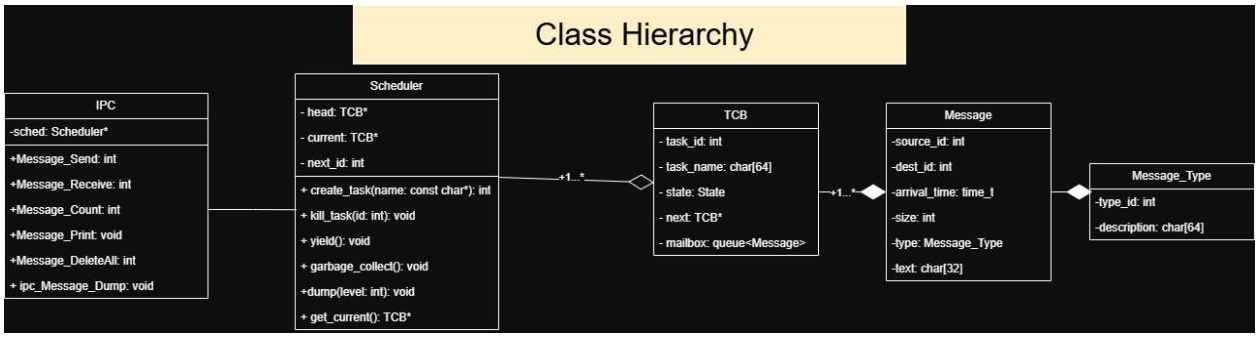
Design Diagrams:

The data structure of the message system uses the TCB where there is a task name, id number, state, and a mailbox. Each TCB is linked to each other and each has a pointer that points to the head and the next task. The TCB also has a mailbox. The mailbox is a queue data structure, using FIFO. For the mailbox, each message uses a struct that stores data for the messages' destination, content, source, time arrival, and size.

The ICP class and the Scheduler have an associative relationship; ICP uses Scheduler to find and use tasks. The scheduler has an aggregation relationship with the TCB. Each TCB has a composition relationship with the Messages, as TCB controls the life of the mailbox for each message. The Messages also has a composition relationship with Message_Type. The type is required for the structure of the message.